ARTIFICIAL INTELLIGENCE

End term Assignment

1 FOUNDATIONS OF AI

1.1 THE TURING TEST AS A BENCHMARK

The Turing Test evaluates whether a machine can imitate human conversation well enough that a human judge cannot tell the difference. If a machine passes this test, it shows that the system can generate responses that match human linguistic behavior and reasoning patterns (Russell & Norvig, 2021). Passing the test does not prove that the machine "thinks" like a human, but it shows that it can understand prompts, respond appropriately, and maintain context in a conversation. For example, a chatbot that handles customer queries smoothly without being identified as a bot demonstrates this level of capability.

1.2 CHARACTERISTICS OF AN AUTONOMOUS INTELLIGENT AGENT

An intelligent agent with full autonomy can sense its environment, make independent decisions, learn from experience, and act without step-by-step instructions (Poole & Mackworth, 2017). A self-driving car is a practical example because it observes the road, predicts hazards, and adjusts speed or direction on its own.

A basic automated program follows fixed rules without learning or adapting. A simple Excel macro is an example. It performs the same actions every time and depends entirely on predefined instructions.

2 Problem-Solving and Search Techniques

2.1 BFS and DFS Mechanics

Breadth-First Search explores graph level by level. It starts at the root and checks all neighbors before moving deeper. Its main strength is its ability to guarantee the shortest solution path when all step costs are equal (Russell & Norvig, 2021).

Depth-First Search moves down one path as far as possible before backtracking. Its main strength is its low memory requirement, which makes it suitable for deep search spaces.

2.2 Role of Heuristics and Drawback of Hill Climbing

Heuristics guide a search algorithm toward promising solutions by estimating how close a current state is to the goal. In Hill Climbing, heuristics help the algorithm move uphill toward a better solution instead of exploring blindly like uninformed methods.

One major drawback is the risk of getting stuck in a local maximum. For example, a company optimizing delivery routes may find a route that seems efficient but fails to reach the truly optimal route because the algorithm stops improving once it reaches a local peak (Arel et al., 2010).

3 Knowledge and Reasoning

3.1 Critical Types of Knowledge

An AI system needs several forms of knowledge to function effectively:

1. **Declarative knowledge** – factual information such as product prices or regulations.
2. **Procedural knowledge** – instructions or rules on how to act, such as how to process an invoice.
3. **Meta-knowledge** – knowledge about when and how to use certain strategies, such as choosing the right financial model in different market conditions (Nilsson, 2009).

These types help an AI system respond correctly within complex, changing environments.

3.2 Monotonic vs. Non-Monotonic Reasoning

Monotonic reasoning states that once something is true, it remains true even after adding new information. Non-monotonic reasoning allows conclusions to change when new evidence appears.

A business example is credit approval. A customer may qualify initially, but if new information shows a sudden drop in income, the system needs to reverse its earlier conclusion. This flexibility is essential in risk-sensitive environments (Brachman & Levesque, 2004).

4 Logic and Learning

4.1 Inference with $P \rightarrow Q$ and $\neg Q$

Given the statements $P \rightarrow Q$ and $\neg Q$, we can use Modus Tollens to conclude $\neg P$. If P were true, Q must follow. Since Q is false, P cannot be true. This rule helps AI systems make consistent logical deductions.

4.2 How a Decision Tree Classifies Data

A decision tree classifies data by repeatedly splitting the dataset into branches based on features that best separate the classes. At each node, the algorithm chooses the feature that provides the highest information gain or lowest impurity (Quinlan, 1993). The goal is to create pure leaf nodes where most or all items belong to the same class. For example, a customer churn model may split data first by contract type, then payment method, and then by customer tenure.

5 AI Applications and Implications

5.1 Components of an Expert System

An expert system includes a knowledge base, an inference engine, and a user interface. The knowledge base stores expert-level facts and rules. The inference engine applies those rules to draw conclusions, making it the core decision-making component. It reasons through the rules, evaluates conditions, and produces recommendations, similar to how a human specialist thinks through a problem (Durkin, 1994).

5.2 Importance of AI Ethics for Business Leaders

Business leaders need a solid understanding of AI ethics because their decisions affect both customers and society. First, ethical awareness helps prevent harmful outcomes such as discrimination or privacy violations, which can damage trust and reputation. Second, leaders must understand the broader societal implications of automation, data usage, and algorithmic decision-making so they can implement AI responsibly and comply with emerging regulations (Jobin et al., 2019).

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